

Ethan Arnold NE533 Exam 3

100

1) $\alpha + \eta = 10 \times 10^{-6}$ $\dot{F} = 8 \times 10^5 \text{ atoms/m}^2 \text{ s}$ $T_f = 1500 \text{ K}$, $T_{ref} = 300 \text{ K}$
 $\Delta p_0 = 0.015$, $\beta_0 = 5 \text{ mV/kV}$, $\rho = 10.97 \text{ g/cc}$, $t = 300 \text{ days}$
 $6\gamma = \alpha \Delta T = 10 \times 10^{-6} \frac{1}{K} (1500 \text{ K} - 300 \text{ K})$
 $t_T = 0.012$

$t_D = \Delta p_0 \left(e^{\frac{\beta \ln 6.01}{\text{COPP}}} - 1 \right)$, $T_f = 1500 \text{ K} > 750^\circ \text{C}$ $C_0 = 1$

$\beta_0 = \frac{5 \text{ mV}}{k_B V} \left(\frac{1}{\Delta T} \right) = 0.0053 \text{ FIMA}$

$10.97 \frac{\text{g}}{\text{cc}} \times \frac{1 \text{ mol}}{270 \text{ g}} \times \frac{6.022 \times 10^{23} \text{ atoms}}{1 \text{ mol}} \times \frac{1 \text{ V}}{1 \text{ V}_0} = 2.447 \times 10^{22} \frac{\text{V}}{\text{cc}} = N_V$

$\beta = \frac{\dot{F} t}{N_V} = \frac{(8 \times 10^5 \text{ atoms/m}^2 \text{ s}) (300 \text{ days} \times 24 \text{ hrs} \times 3600 \text{ s})}{2.447 \times 10^{22} \frac{\text{V}}{\text{cc}}} = 0.0847$

since $\beta > \beta_0$, $t_D = \Delta p_0 = 0.015$

$t_{JFP} = 5.577 \times 10^{-2} \rho \beta = 5.577 \times 10^{-2} (10.97 \text{ g/cc}) (0.0847) = 0.052$

$t_{GFP} = 1.96 \times 10^{-8} \rho \beta (2800 - T)^{11.73} x e^{-0.0162(2800 - T)} - 1.78 \rho \beta$, $\rho = 10.97 \text{ g/cc}$, $\beta = 0.0847$, $T = 1500 \text{ K}$
 $t_{GFP} = 2.84 \times 10^{-8}$

$t_{tot} = t_T + t_D + t_{JFP} + t_{GFP} = 0.012 + 0.015 + 0.052 + 2.84 \times 10^{-8} = 0.049$ or **4.9%**

2. $\sigma_m = 60 \text{ MPa}$, $T = 600 \text{ K}$, $LHR = 250 \frac{\text{h}}{\text{cm}}$, $t = 200 \text{ days}$, $A_0 = 3.14 \times 10^{24}$
 $B = 4.2519 \times 10^{10} - 2.2185 \times 10^7 T$, $n = 5$ $Q = 2.7 \times 10^5$, $C_0 = 1.654 \text{ E}^{-24}$, $C_1 = 0.85$
 $C_2 = 1$

$B = 4.2519 \times 10^{10} - 2.2185 \times 10^7 (600) = 2.92 \times 10^{10}$

$t_{in} = (3.14 \times 10^{24}) \left(\frac{60 \times 10^6}{2.92 \times 10^{10}} \right)^5 e^{(-2.7 \times 10^5 / (8.314 \times 600))}$

$t_{in} = 3.584 \times 10^{13} \frac{1}{s}$

$t_{cr} = (1.654 \times 10^{-24}) (3 \times 10^{11} (250))^{0.85} (60)^1 = 6.17 \times 10^{-11} \frac{1}{s}$

$t_{cr} = [(3.584 \times 10^{13} \frac{1}{s}) + 6.17 \times 10^{-11} \frac{1}{s}] \times 200 \times 24 \times 3600 = 0.0011$ or **0.11%**

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3. $t = 250$ days $T = 1400$ K $\dot{F} = 2 \times 10^{13} \frac{1}{\text{cm}^2 \cdot \text{s}}$, $r = 5$ micron, $r = 40$ mm
 $\dot{F}$ implies in-pile

$$5 \mu\text{m} \left(\frac{1e^{-4}}{1 \mu\text{m}} \right) = 5e^{-4} \text{ cm and } 40e^{-4} \text{ cm}$$

$$D = D_1 + D_2 + D_3$$

$$D_1 = 7.6 \times 10^{-6} e^{\left(\frac{-3.03}{18.617e^{-5} (1400\text{K})} \right)} = 9.39e^{-17} \frac{\text{cm}^2}{\text{s}}$$

$$D_2 = 1.41 \times 10^{-16} e^{\left(\frac{-1.61}{18.617e^{-5} (1400\text{K})} \right)} \sqrt{2 \times 10^{13}} = 3.279 \times 10^{-16} \frac{\text{cm}^2}{\text{s}}$$

$$D_3 = 2.0 \times 10^{-30} (2 \times 10^{13}) = 4e^{-17} \frac{\text{cm}^2}{\text{s}}$$

$$D = 9.39e^{-17} + 3.279e^{-16} + 4e^{-17} = 4.618e^{-16}$$

$$\tau = \frac{D \cdot t}{a^2} = \frac{(4.618e^{-16} \text{ cm}^2) (250 \times 24 \times 3600)}{(5e^{-4})^2} = 0.0399$$

$$\tau = \frac{4.618e^{-16} \times 250 \times 24 \times 3600}{(40e^{-4})^2} \rightarrow \tau = 0.000623$$

$$\frac{1}{\pi^2} = 0.1 \text{ so both } < \frac{1}{\pi^2}$$

$$f_5 = 4 \sqrt{\frac{0.0399}{\pi}} - \frac{3}{2} (0.0399) = 0.39 \leftarrow \text{seems high}$$

$$f_{40} = 4 \sqrt{\frac{0.000623}{\pi}} - \frac{3}{2} (0.000623) = 0.055$$

4. The typical charge state of V in VO_2 is V^{4+} . It can change due to ratios in defects or fission products with a different charge like Sr^{2+} or O/M ratio (excess oxygen).

5. Oxygen concentration is somewhat stable at the mol/mol ratio. Molybdenum acts as a buffer for O_2 and allows a determination of the O/M ratio during PIE. O_2 does decrease near the cladding as it enters the cladding.

6. melting temperature, thermal conductivity, diffusion properties (creep, FGR)

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7. Soluble oxides - increase swelling and decrease thermal conductivity
Insoluble oxides - increase swelling and decrease thermal conductivity
Metal (precipitates) - increase thermal conductivity
Volatiles - gas near pellet center, solid near pellet surface, cause fuel swelling and corrode cladding
Noble gases - cause fuel swelling due to pores and decrease thermal conductivity when released to the gap
8. Stage 1: Fission gas formation in the grain and diffusion to grain boundary. Fission gas forms in a grain and diffuses to the boundary if it is not trapped in small pores. Stage 2: nucleation and interconnection. Fission gas nucleates at the boundary and interconnects to form larger bubbles. Stage 3: release to free volume. Fission gas transports along grain boundaries until it is released into the free volume.
9. Creep is a method of plastic deformation driven by defect diffusion despite having an applied stress below the yield stress. One mechanism is dislocation creep where dislocations glide and/or climb over defects while preferentially absorbing interstitials. This type of creep is known as power law creep and occurs at higher stresses and temperatures.
10. Irradiation growth occurs at point defects nucleate and preferentially form interstitial loops along pyramidal planes. There is an initial spurt of growth and then these interstitial loops slowly allow growth pyramidally over time. Eventually vacancies preferentially form loops on the basal planes leading to rapid, breakaway growth.

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12. Early in life, higher system pressure forces the cladding to creep down. Eventually, the rod creep and fuel swelling cause pellet/clad contact and close the gap. During this time, the cladding usually lengthens. After contact, the fuel pushes the cladding out in some areas causing local axial tensions. Some tube elongation may still occur when a significant portion of the cladding is in contact with the fuel. Expansion of the cladding from creep and the stress provided by the pellet begins again and the cladding may shorten.

13. 1) A corrosive environment - SCC requires a corrosive environment. The gaseous, volatile fission products (I_2 , I) released from the pellet may travel down thermal gradients along fuel cracks where they can deposit and corrode cladding locally forming ZrI_4 provided there is some penetration of the oxide layer.

2) Susceptible material - SCC requires a susceptible material. Some work has been done to alloy Zr with metals to decrease corrosion (Nb , Sn , Cu), but all Zr alloys are still susceptible to corrosive fission products.

3) Sufficient stress - SCC requires sufficient stress to occur. Generally, there is radial stress in the cladding and hoop stress.

Things like missing pellet surfaces can cause dramatic stress concentrations making the cladding much more susceptible to PCI.

4) Sufficient time - Sufficient time must occur for SCC to propagate. ZrI_4 must transport into the cracks, oxidize to create ZrO_2 and leave I_2 free near the crack tips to propagate. A corrosive environment must form, initiation of a crack must occur, propagation must occur, and then failure happens.

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5/5

Iodine is released with Cesium in the form of CsI that transports down thermal gradients in pellet cracks to the cladding/gap. CsI then reacts with Zr cladding to form ZrI_4 which can travel down a SiC crack in the cladding. The ZrI_4 then oxidizes turning into ZrO_2 which passivates the crack edges forcing the volatile corrosive Iodine to the crack tip. The process may then repeat with ZrI_4 forming, changing to ZrO_2 , and releasing I_2 back to the crack tip.

10. The high burnup structure is a 20% porous structure formed due to polygonization ~~restructuring~~ as it is more energetically favorable. Dislocations rearrange into subgrains with misaligned grains. The HBS traps fission gas and has better kth. It forms due to high local burnup from resonance capture near the pellet rim.